UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 014 · Applications of LDEs: LCR circuits · compiled 17 October 2026

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1 Short notes

1 · Modelling the series LCR circuit

Series LCR circuit: drops $L\frac{dI}{dt}$, RI, $\frac{Q}{C}$; Kirchhoff's voltage law: their sum is $E(t)$. With $I=\frac{dQ}{dt}$: $LQ''+RQ'+\frac{Q}{C}=E(t)$, a second-order linear DE with constant coefficients. Units: H, Ω, F, V, A, C. Conditions $Q(0)$, $I(0)=Q'(0)$. Differentiating gives $LI''+RI'+\frac{I}{C}=E'(t)$ with $LI'(0)=E(0)-RI(0)-\frac{Q(0)}{C}$.

2 · Solving the circuit equation: CF, PI and damping

Divide by L; AE $Lm^2+Rm+\frac{1}{C}=0$. $R^2>\frac{4L}{C}$ overdamped, $=$ critical, $<$ underdamped ($e^{-Rt/2L}(c_1\cos\omega_d t+c_2\sin\omega_d t)$). CF = transient, PI = steady state; battery: PI $=CE$. Slide example ($R=8$, $L=2$, $C=0.1$, $E=10$): $Q=1-e^{-2t}(\cos t+2\sin t)$, $I=5e^{-2t}\sin t$.

3 · Past-paper circuits: AC sources and free discharge

ESA s3: $Q''+40Q'+625Q=100\cos 5t$, $Q=e^{-20t}(c_1\cos 15t+c_2\sin 15t)+0.15\cos 5t+0.05\sin 5t$. ESA s4: $Q''+20Q'+500Q=12\sin 10t$, $Q=e^{-10t}(c_1\cos 20t+c_2\sin 20t)+0.024\sin 10t-0.012\cos 10t$. PI by $D^2\to-\omega^2$ and rationalising. CW P36: $q=e^{-500t}(0.002\cos 500\sqrt{7}t+0.000756\sin 500\sqrt{7}t)$.

4 · First-order circuits: RL and RC

RL: $Li'+Ri=E\sin\omega t$, $i(0)=0$: $i=\frac{E}{\sqrt{R^2+L^2\omega^2}}[\sin(\omega t-\varphi)+e^{-Rt/L}\sin\varphi]$, $\tan\varphi=\frac{L\omega}{R}$. RC: $Ri'+\frac{i}{C}=E'$, $E=E_0\cos pt$: $i=\frac{CpE_0}{1+p^2R^2C^2}[RCp\cos pt-\sin pt-pRCe^{-t/RC}]$.

2 Keyword index

22 terms; the number in brackets is the topic.

CE (2). Final charge for a battery.

$I = \frac{dQ}{dt}$ (1). Current is the rate of change of charge.

Critically damped (2). $R^2 = \frac{4L}{C}$.

Current equation (1). $LI'' + RI' + \frac{I}{C} = E'(t)$.

Damped frequency (2). $\omega_d = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$.

Free discharge (3). $E = 0$; charge = CF only.

Impedance (2). $Z = \sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$.

Initial conditions (1). $Q(0)$ and $I(0)$.

Kirchhoff's voltage law (1). Sum of drops = EMF.

LCR (RLC) circuit (1). L, R, C and $E(t)$ in series.

LCR equation (1). $LQ'' + RQ' + \frac{Q}{C} = E(t)$.

Ohm's law (1). $E_R = RI$.

Overdamped (2). $R^2 > \frac{4L}{C}$.

Phase lag (4). $\varphi = \tan^{-1}\frac{L\omega}{R}$.

RC circuit (4). $Ri' + \frac{i}{C} = E'$; transient $e^{-t/RC}$.

Resonance (2). Largest response near $\omega = \frac{1}{\sqrt{LC}}$.

RL circuit (4). $Li' + Ri = E$; transient $e^{-Rt/L}$.

Steady state (2). PI; remains.

Time constant (4). $\frac{L}{R}$ or RC.

Transient (2). CF; decays.

Underdamped (2). $R^2 < \frac{4L}{C}$, decaying oscillation.

Voltage drop (1). $L\frac{dI}{dt}$, RI, $\frac{Q}{C}$.

3 Rapid recall

1	Voltage drop across an inductor?	$L\frac{dI}{dt}$.
2	Across a capacitor?	$\frac{Q}{C}$.
3	Kirchhoff's voltage law for the series circuit?	$L\frac{dI}{dt} + RI + \frac{Q}{C} = E(t)$.
4	The LCR equation in Q?	$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$.
5	Its order and type?	Second-order linear, constant coefficients, non-homogeneous.
6	How many initial conditions, and which?	Two: $Q(0)$ and $I(0) = Q'(0)$.
7	Overdamped when?	$R^2 > \frac{4L}{C}$.
8	Underdamped transient?	$e^{-Rt/2L}(c_1\cos\omega_d t + c_2\sin\omega_d t)$.
9	PI for a battery E?	CE (charge); the current tends to 0.
10	Slide example $R=8$, $L=2$, $C=0.1$, $E=10$: $I(t)$?	$5e^{-2t}\sin t$.
11	...and $Q(t)$?	$1 - e^{-2t}(\cos t + 2\sin t)$.
12	ESA $R=40$, $L=1$, $C=16\times10^{-4}$, $E=100\cos 5t$: roots?	$-20 \pm 15i$.
13	...its steady-state charge?	$0.15\cos 5t + 0.05\sin 5t$.
14	ESA $R=20$, $L=1$, $C=0.002$, $E=12\sin 10t$: roots?	$-10 \pm 20i$.
15	...its steady-state charge?	$0.024\sin 10t - 0.012\cos 10t$.
16	CW P36 damped frequency?	$500\sqrt{7} \approx 1322.9$ rad/s.
17	RL circuit transient?	$e^{-Rt/L}$.
18	RC circuit equation in i?	$R\frac{di}{dt} + \frac{i}{C} = \frac{dE}{dt}$.

Mnemonic (study aid). *"L-D-I, R-I, Q over C."* The three drops in the order of the slide: $L\frac{dI}{dt}$, RI, $\frac{Q}{C}$ (study aid).

4 Last 10 minutes before the exam

Top facts

1. $LQ'' + RQ' + \frac{Q}{C} = E(t)$, $I = \frac{dQ}{dt}$; divide by L (and E too) before the AE.
2. $Q(0)$ gives c_1; differentiate Q, then $I(0)$ gives c_2.
3. $R^2 \gtrless \frac{4L}{C}$: over / critical / underdamped; the CF always decays (transient), the PI stays (steady state).
4. Battery: PI $= CE$. Slide: $Q = 1 - e^{-2t}(\cos t + 2\sin t)$, $I = 5e^{-2t}\sin t$.
5. ESA s3: $Q = e^{-20t}(c_1 \cos 15t + c_2 \sin 15t) + 0.15\cos 5t + 0.05 \sin 5t$.
6. ESA s4: $Q = e^{-10t}(c_1 \cos 20t + c_2 \sin 20t) + 0.024 \sin 10t - 0.012 \cos 10t$.
7. No initial conditions in the question: keep c_1, c_2.
8. RL: $i = \frac{E}{\sqrt{R^2 + L^2\omega^2}}[\sin(\omega t - \varphi) + e^{-Rt/L}\sin\varphi]$; RC: transient $e^{-t/RC}$.

Top formulas

E1.2 · Kirchhoff's voltage law

ESA ★★

$$E(t) = V_L + V_R + V_C = L\frac{dI}{dt} + RI + \frac{Q}{C}$$

GIVES	The circuit equation in I and Q.
APPLIES	Series circuit.
NOTE	Sum of the drops = the EMF supplied.

S014·5 · S014·6 · T014 00:01:51 · T014 00:02:21

E1.3 · The LCR equation for the charge

ESA ★★

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t), \qquad I = \frac{dQ}{dt}$$

GIVES	A second-order linear DE with constant coefficients for $Q(t)$.
APPLIES	Series LCR circuit; L, R, C constant.
NOTE	Divide by L before finding the AE: $Q'' + \frac{R}{L}Q' + \frac{1}{LC}Q = \frac{E}{L}$.

S014·6 · S014·7 · S014·14 · T014 00:02:58 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b

E1.4 · Initial conditions

ESA ★★

$$Q(0) = Q_0, \qquad I(0) = Q'(0) = I_0$$

GIVES	The two conditions that fix c_1, c_2.
APPLIES	Usually $Q_0 = I_0 = 0$ (switch closed at $t = 0$).
NOTE	Use $Q(0)$ in $Q(t)$, then differentiate $Q(t)$ and use $I(0)$.

S014·7 · S014·11 · S014·12 · T014 00:03:41 · T014 00:08:21

E2.1 · Auxiliary equation and its roots

ESA ★★

$$Lm^2 + Rm + \frac{1}{C} = 0, \qquad m = \frac{-R \pm \sqrt{R^2 - 4L/C}}{2L}$$

GIVES The CF (transient) of the charge.

APPLIES Every LCR problem.

NOTE Slide example: $2m^2 + 8m + 10 = 0$, $m = -2 \pm i$.

S014·9 · S014·15 · T014 00:06:48

E2.2 · Damping regimes

ESA ★

$$R^2 > \tfrac{4L}{C} : \; Q_{tr} = c_1 e^{m_1 t} + c_2 e^{m_2 t}$$
$$R^2 = \tfrac{4L}{C} : \; Q_{tr} = (c_1 + c_2 t) e^{-Rt/(2L)}$$
$$R^2 < \tfrac{4L}{C} : \; Q_{tr} = e^{-Rt/(2L)}(c_1 \cos \omega_d t + c_2 \sin \omega_d t)$$

GIVES The form of the transient.

APPLIES Compare R^2 with $\frac{4L}{C}$.

NOTE Over-, critically, underdamped. All decay: $m_1, m_2 < 0$ or real part $-\frac{R}{2L} < 0$.

ADDED FOR CLARITY S014·9 · S014·15

E2.4 · Solution = transient + steady state

ESA ★★

$$Q(t) = \underbrace{\text{CF}}_{\text{transient}} + \underbrace{\text{PI}}_{\text{steady state}}, \qquad E = \text{const} \Rightarrow \text{PI} = CE$$

GIVES The structure of every answer.

APPLIES Any $E(t)$.

NOTE Constant E: $\frac{E/L}{D^2 + \frac{R}{L}D + \frac{1}{LC}}$ with $D \to 0$ gives CE (slide: $\frac{5}{5} = 1 = 0.1 \times 10$).

S014·10 · T014 00:07:37 · T014 00:07:52

E4.1 · RL circuit with a sinusoidal source

ESA ★★

$$L\frac{di}{dt} + Ri = E \sin \omega t, \; i(0) = 0 :$$
$$i = \frac{E}{\sqrt{R^2 + L^2\omega^2}} \left[\sin(\omega t - \varphi) + e^{-Rt/L} \sin \varphi \right], \; \varphi = \tan^{-1} \frac{L\omega}{R}$$

GIVES The current in an RL circuit.

APPLIES No capacitor; switch closed at $t = 0$.

NOTE Transient $e^{-Rt/L}$, time constant $\frac{L}{R}$.

CW U2 P34